UNIVERSITY OF BAGHDAD

COLLEGE OF MEDICINE

2026-2027

LECTURE TITLE

Introduction to Diagnostic Imaging & Chest Radiology

Department of Surgery / Radiology Section

Stage 4 • Academic Year 2026–2027

Lecturer Doctor Abdullah Ahmed

College of Medicine – University of Baghdad

Learning Objectives

- Describe the basic principles of diagnostic imaging modalities (X-ray, CT, MRI, ultrasound).
- Identify standard chest X-ray projections and technical requirements for optimal imaging.
- Interpret normal anatomical structures on chest X-ray and distinguish them from common abnormalities.
- Summarize the main indications and advantages of chest CT in clinical practice.
- Recognize key radiological signs of common lung diseases on chest X-ray and CT.

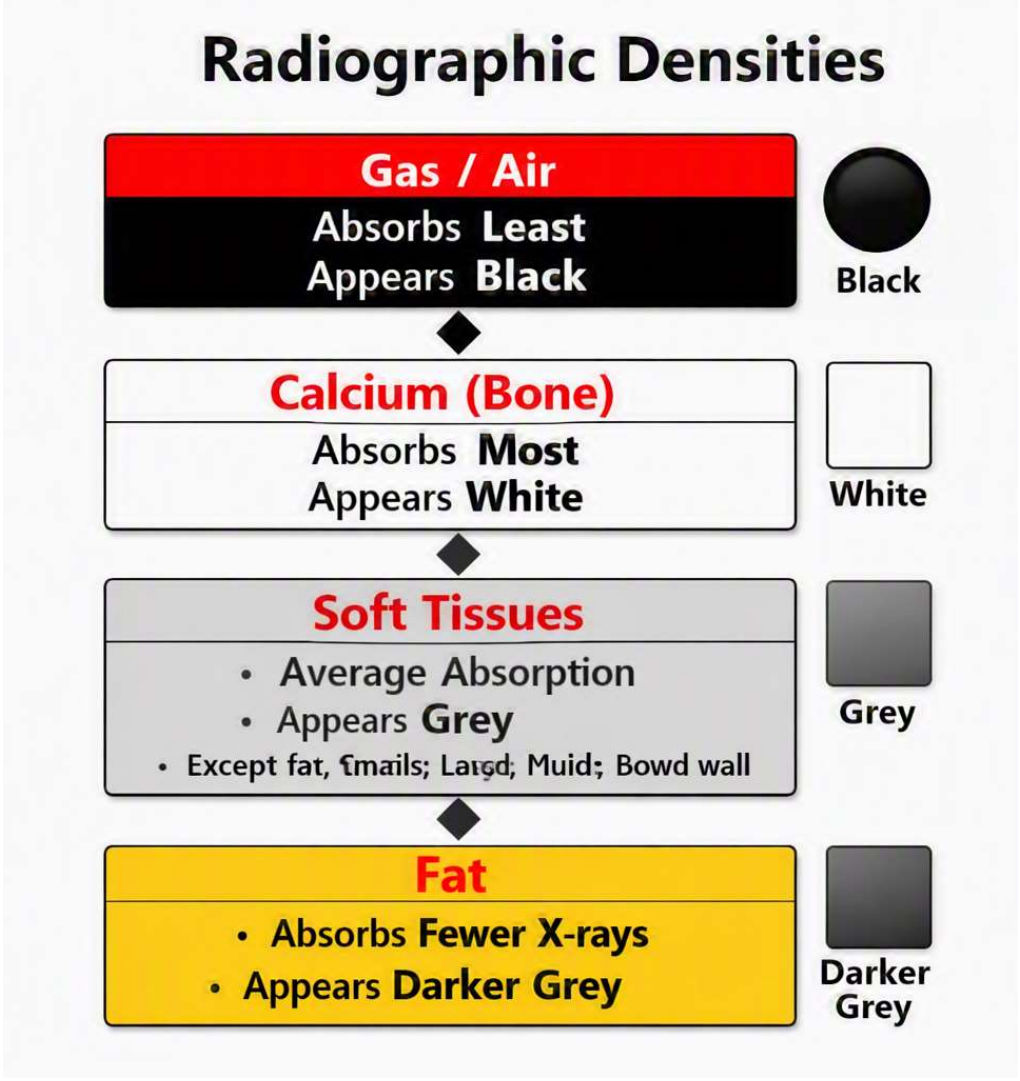
Principles of Radiographic Density

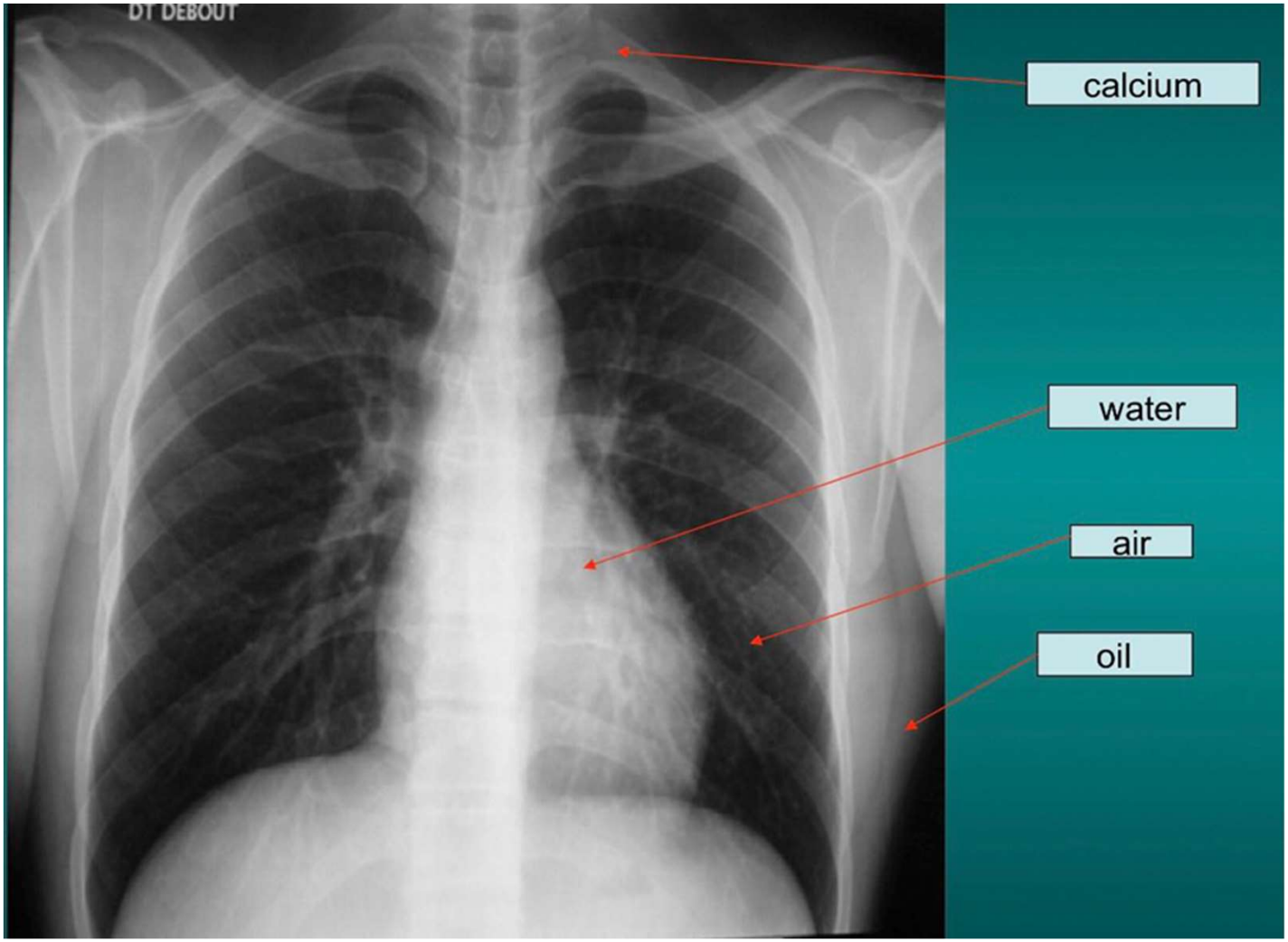
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- **X-rays are absorbed** to a **variable extent** as they pass through the body.
- **Visibility of normal structures and disease** depends on this **differential absorption**.

In conventional radiography, there **are four basic densities**:

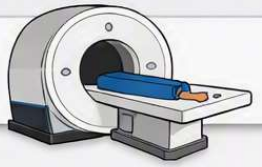
- **Gas**
- **Fat**
- **Soft Tissues**
- **Calcified Structures**


Radiographic Densities
Gas / Air
Absorbs Least
Appears Black
Black
Calcium (Bone)
Absorbs Most
Appears White
White
Soft Tissues
• Average Absorption
• Appears Grey
• Except fat, tmails; Laıgd; Muid; Bowd wall
Grey
Fat
• Absorbs Fewer X-rays
• Appears Darker Grey
Darker Grey


DT DEBOUT
calcium
water
air
oil

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Computed Tomography ((CT) Principles



Utilizes **X-rays** to create images



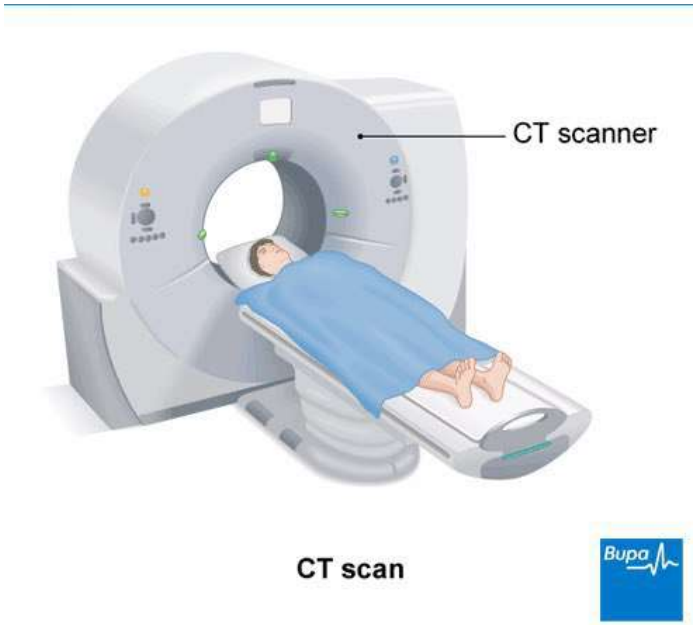
More Sensitive X-ray **Detection** System



Produces **Detailed** **Cross-Sectional** Slices

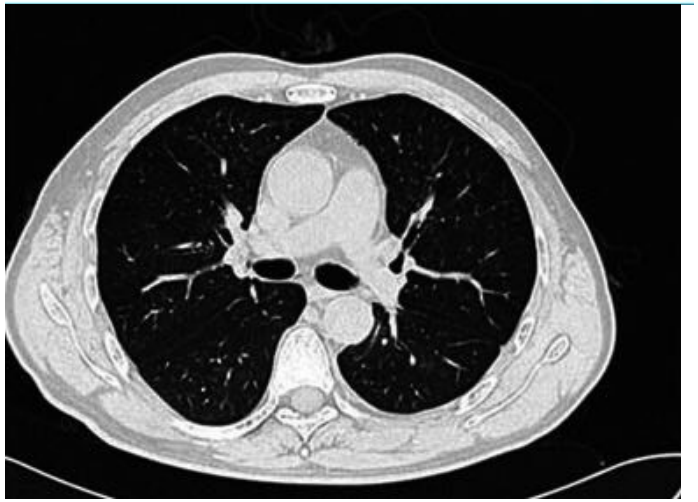


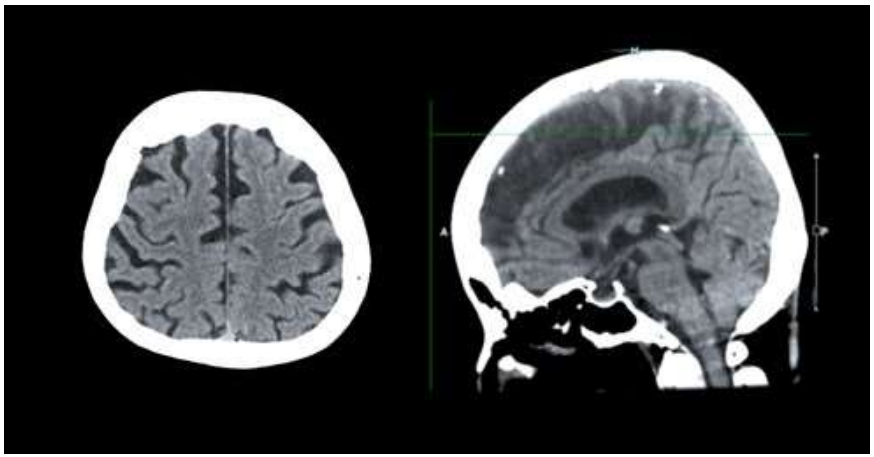
Data Processed by **Computer**



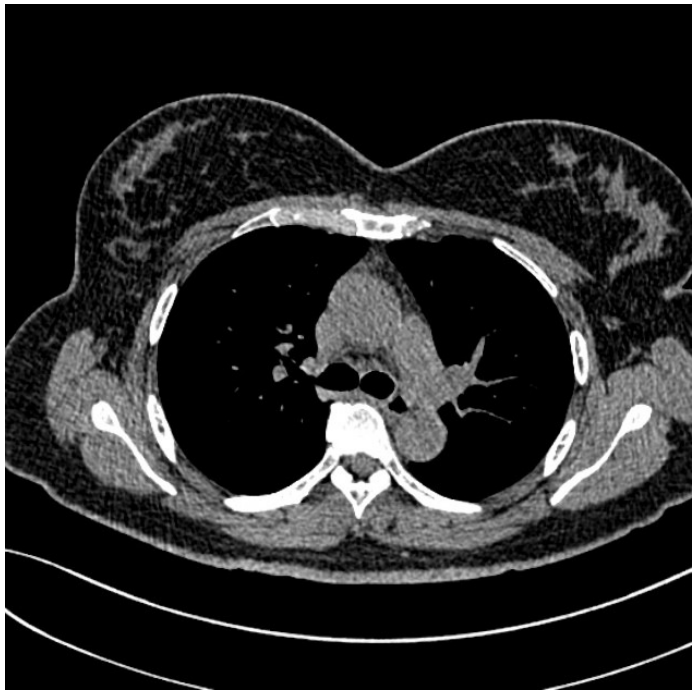
Main Indications:

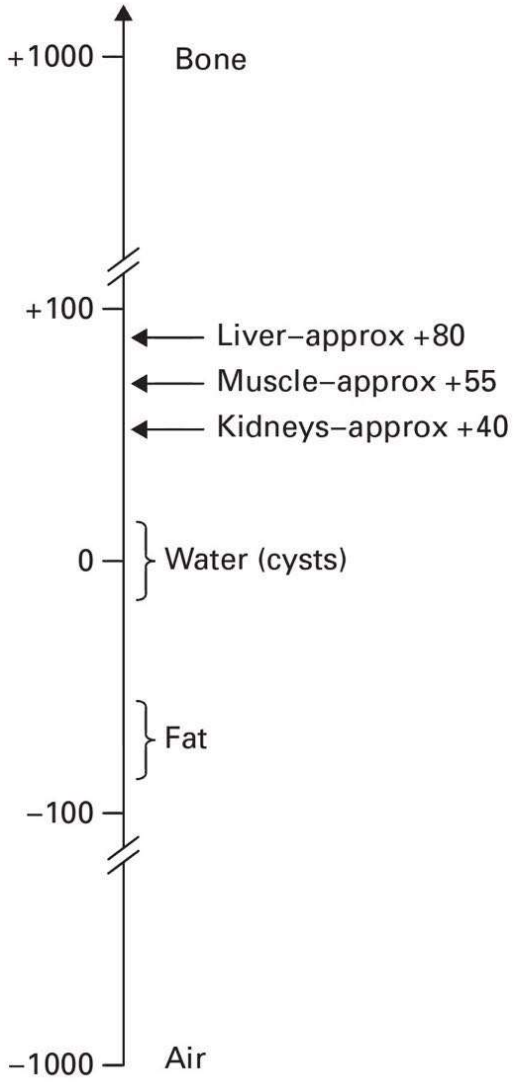
1. Brain and Skull
2. Head and Neck
3. Chest
4. Abdomen
5. Bony Skeleton

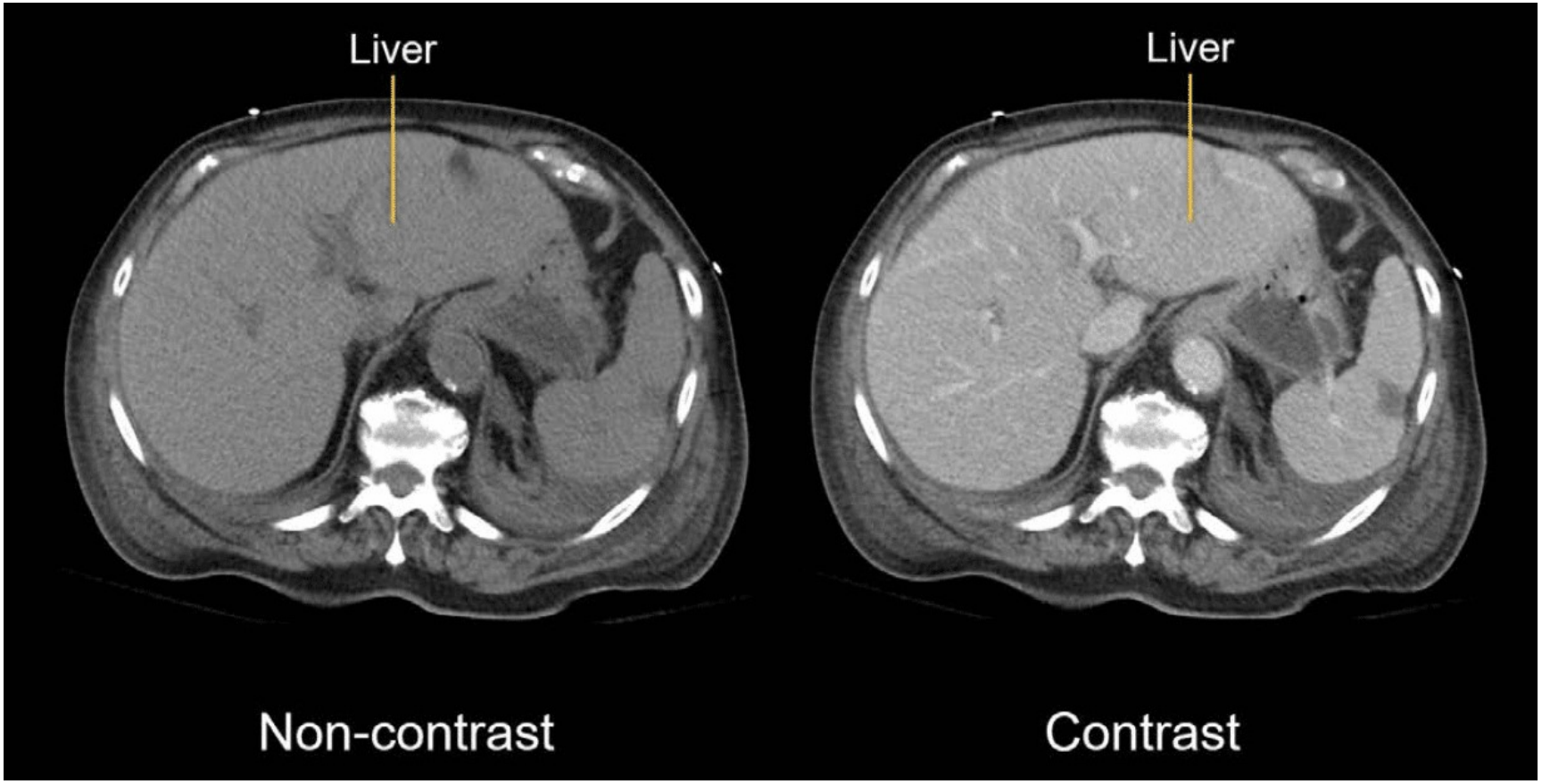


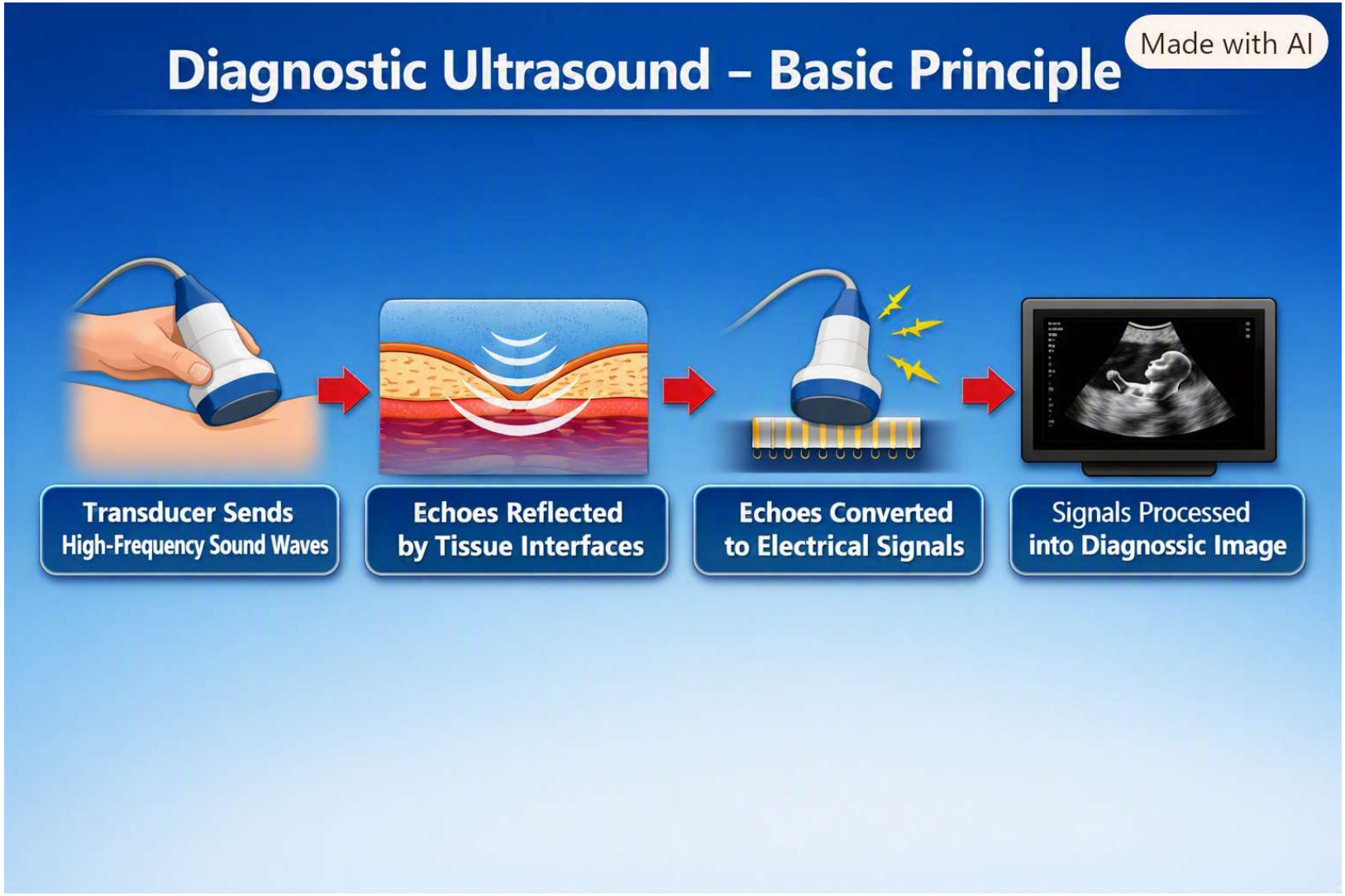


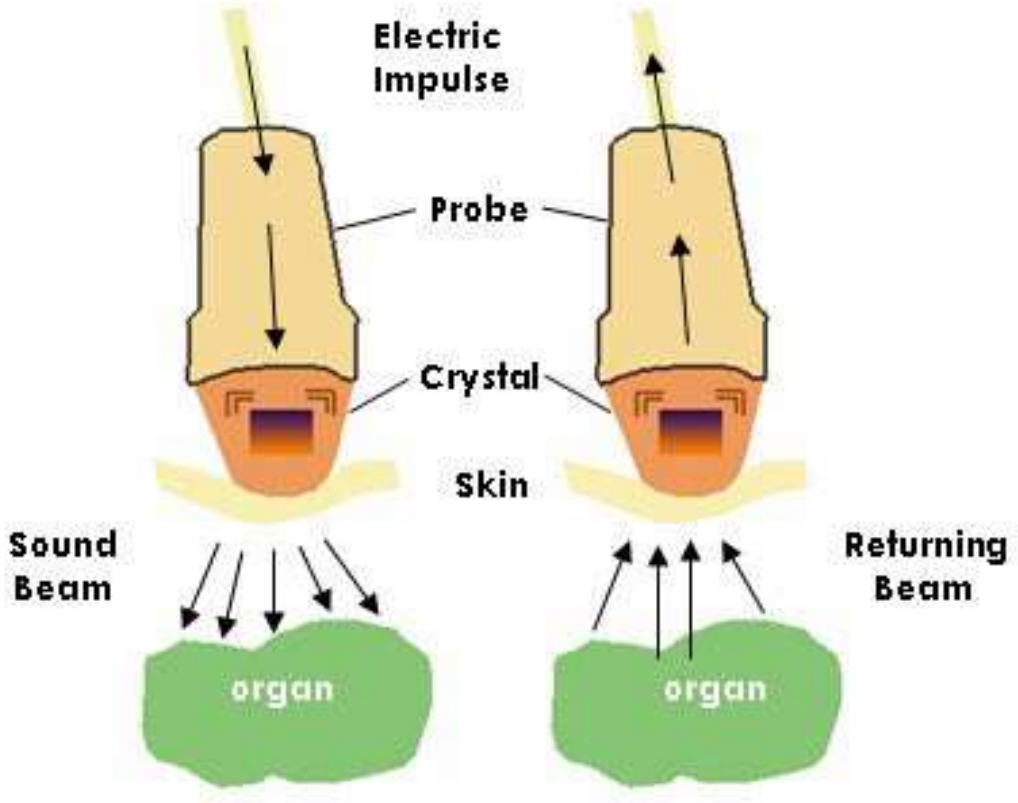
The value of X - ray attenuation is called (**Hounsfield units**)






Liver
Liver
Non-contrast
Contrast


Diagnostic Ultrasound – Basic Principle
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Transducer Sends
High-Frequency Sound Waves
Echoes Reflected
by Tissue Interfaces
Echoes Converted
to Electrical Signals
Signals Processed
into Diagnossic Image

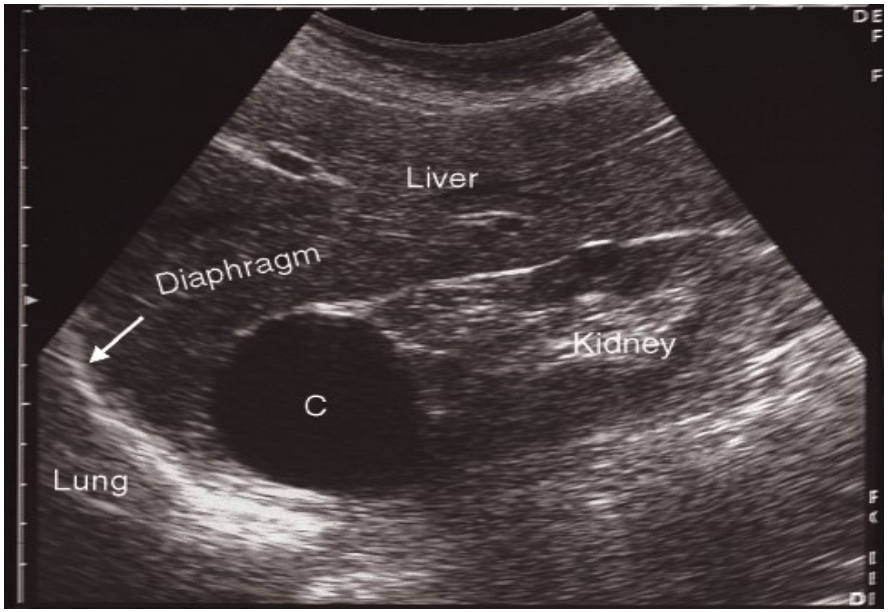

Electric
Impulse
Probe
Crystal
Skin
Sound
Beam
Returning
Beam
organ
organ



Ultrasound is often used to determine whether a structure is **solid or cystic.**

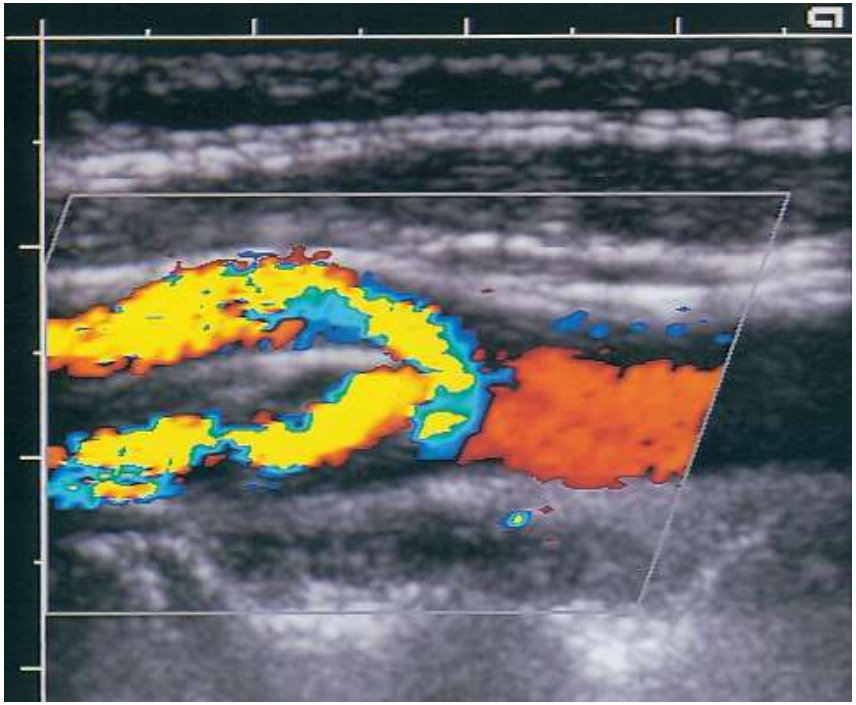
Advantages:

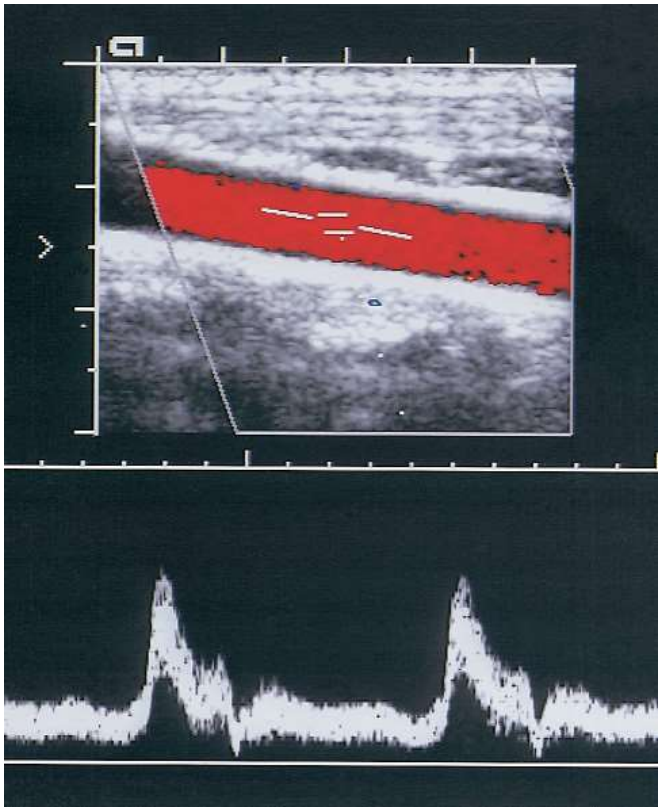
1. **No** radiation exposure
2. Simple and **safe**
3. **Not costly**

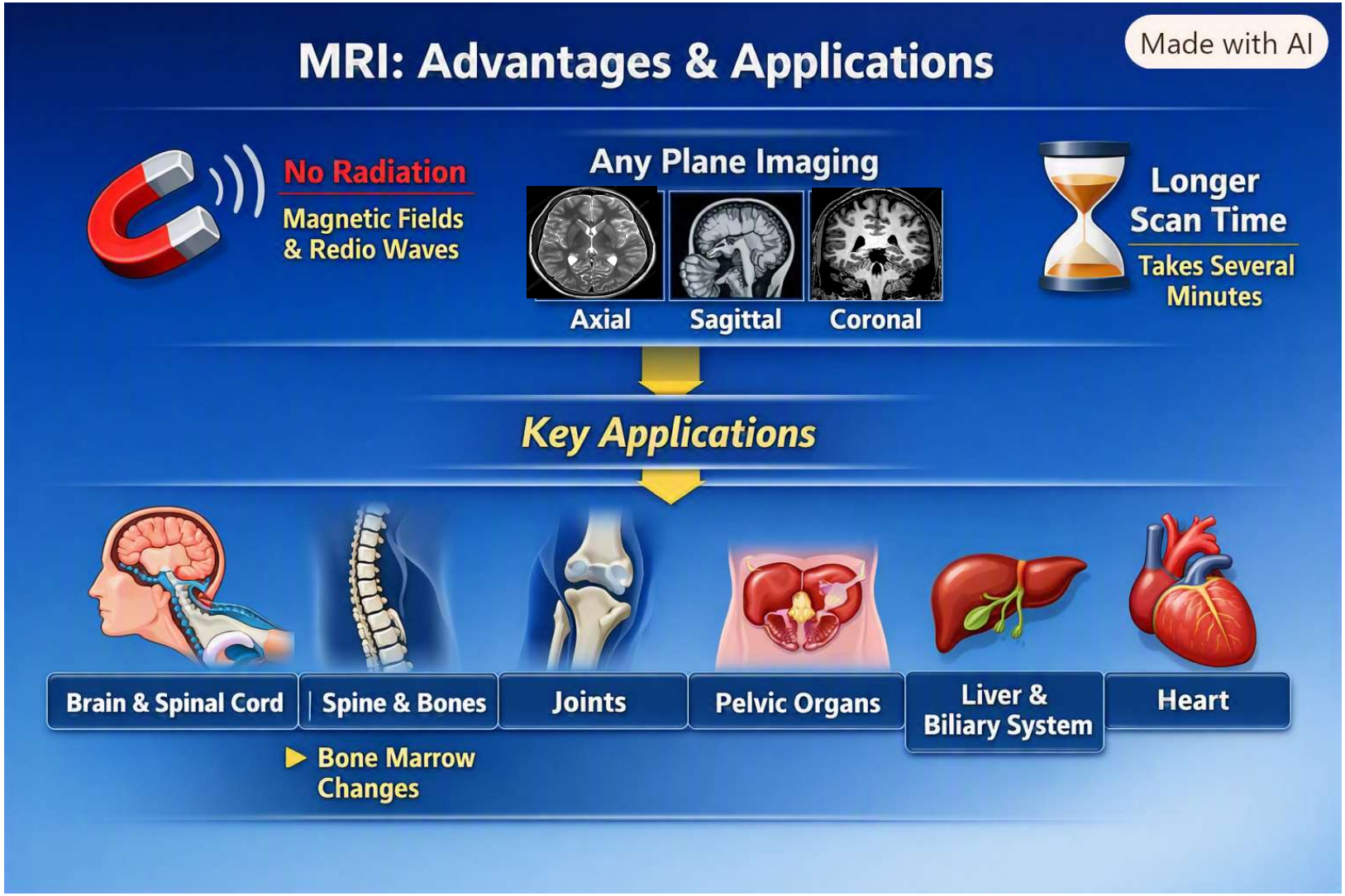


Doppler:

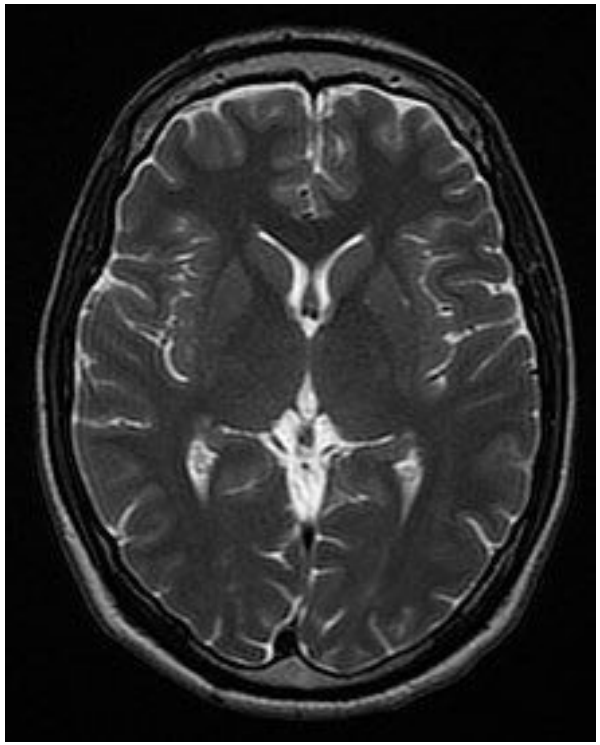
- **Patency** of blood vessels.
- **Velocity** of blood.
- Blood flow **wave form.**






MRI: Advantages & Applications
Made with AI
No Radiation
Magnetic Fields & Redio Waves
Any Plane Imaging
Axial
Sagittal
Coronal
Longer Scan Time
Takes Several Minutes
Key Applications
Brain & Spinal Cord
Spine & Bones
Bone Marrow Changes
Joints
Pelvic Organs
Liver & Biliary System
Heart





Modality	Principle	Advantages	Limitations	Common Uses
X-ray	Uses ionizing radiation; attenuation depends on tissue density	Quick, cheap, widely available	Radiation exposure, limited soft tissue contrast	Chest imaging, bone fractures
CT	Rotating X-ray beam + detectors; attenuation measured in Hounsfield units	Excellent cross-sectional detail, fast, good for lung and bone	Higher radiation dose, less soft tissue contrast than MRI	Pulmonary embolism, mediastinal masses, trauma
MRI	Magnetic field + radiofrequency pulses; signal depends on proton relaxation	No radiation, superior soft tissue contrast, multiplanar	Expensive, longer exam time, contraindications (metal implants, claustrophobia)	Brain, spine, joints, soft tissue tumors
Ultrasound	High-frequency sound waves; reflection depends on tissue density	No radiation, portable, safe, inexpensive	Operator dependent, limited penetration in bone/air	Pleural effusion, abdominal organs, vascular Doppler

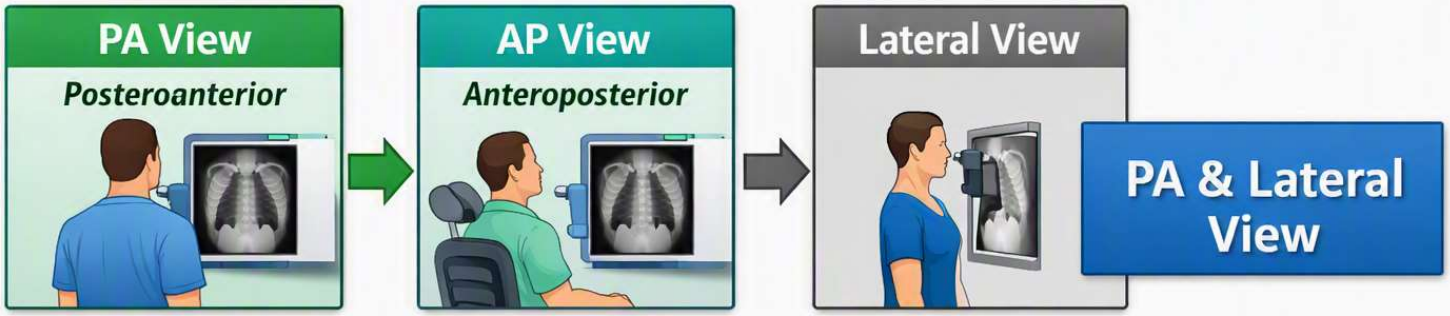
Introduction to Chest Imaging



Chest Radiograph – Standard Views

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Standard Views



Proper Technique



Full Inspiration

Deep Breath In



Upright Position

Patient Standing

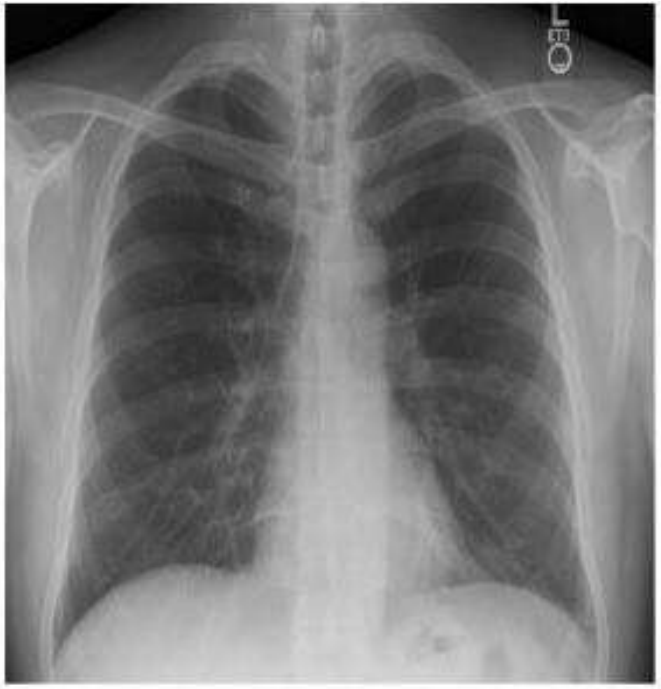


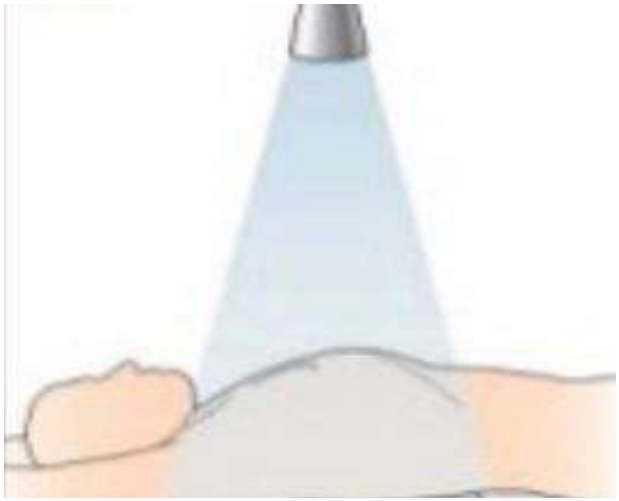
Patient in upright position



PA chest radiograph read as the patient is in front of you

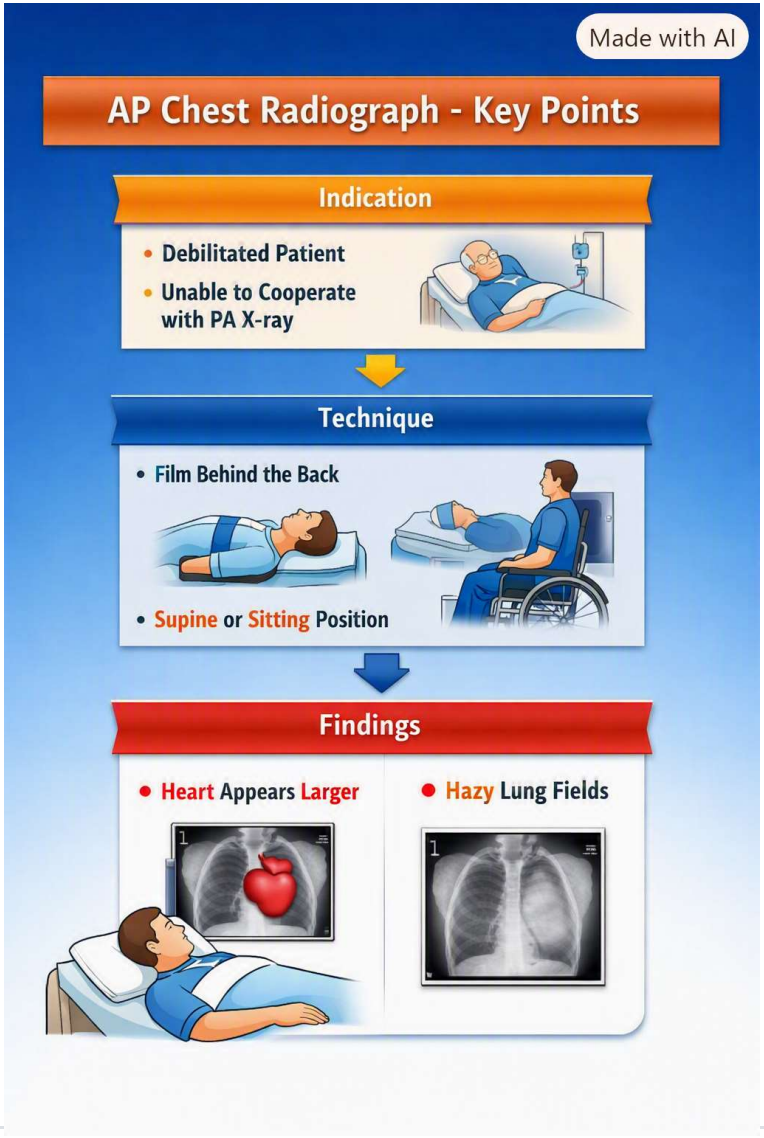




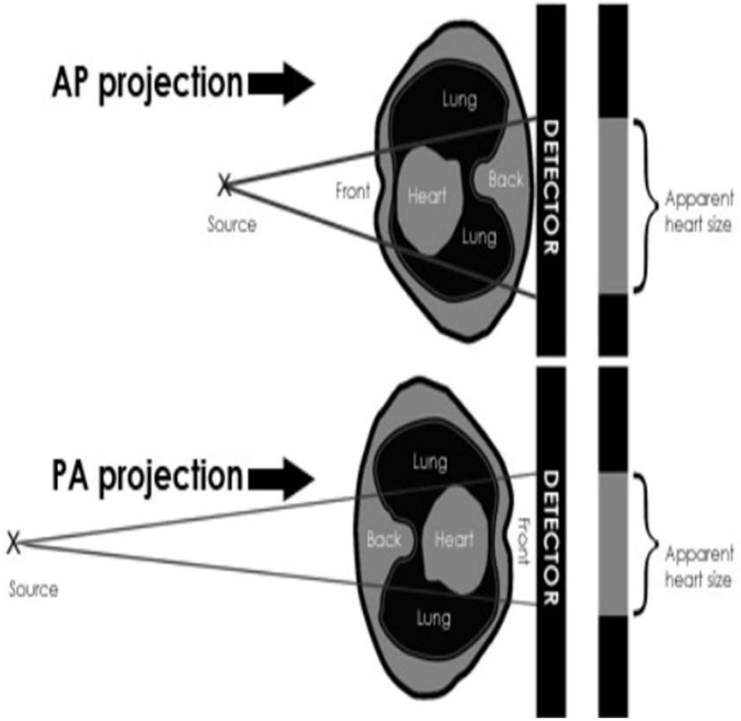







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AP Chest Radiograph - Key Points
Indication
Debilitated Patient
Unable to Cooperate with PA X-ray
Technique
Film Behind the Back
Supine or Sitting Position
Findings
Heart Appears Larger
Hazy Lung Fields

AP v PA projection

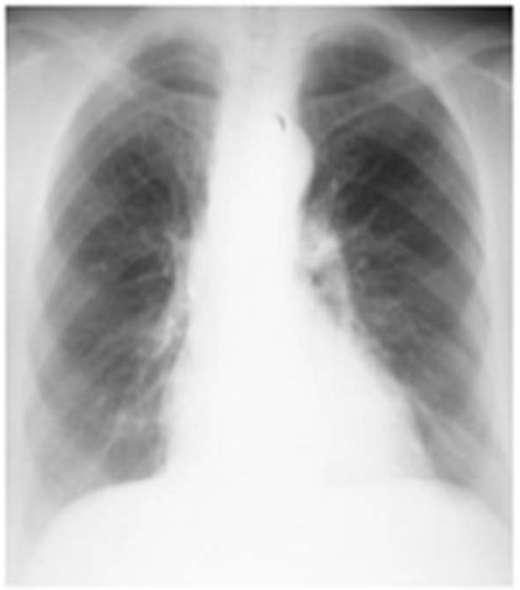


AP v PA projection

- The upper diagram shows an AP projection. Heart size is exaggerated because the heart is relatively farther from the detector, and also because the X-ray beam is more divergent as the source is nearer the patient.
- The lower diagram shows a conventional PA projection. The apparent heart size is nearer to the real size, as the heart is relatively nearer the detector. Magnification of the heart is also minimised by use of a narrower beam, produced by the increased distance between the source and the patient.

Over-penetrated	Under-penetrated
If intervertebral disc are very clearly seen in the film	If intervertebral disc are not seen in the film





Centering

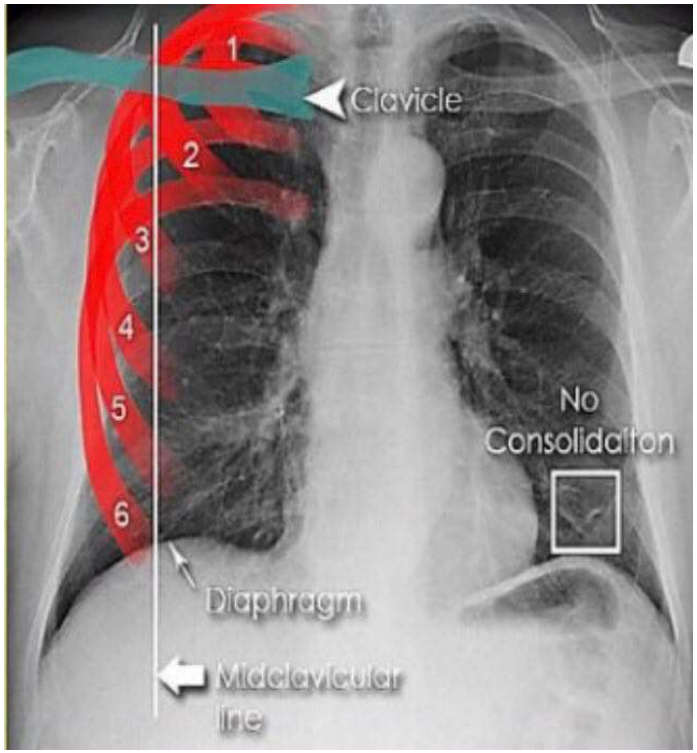
- Medial ends of clavicles should be equidistant from spinous processes of vertebrae.



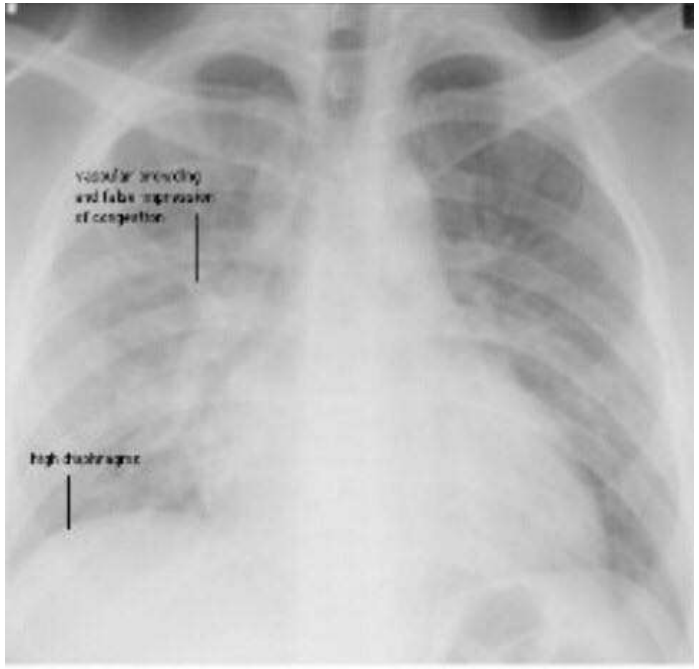
Full inspiration

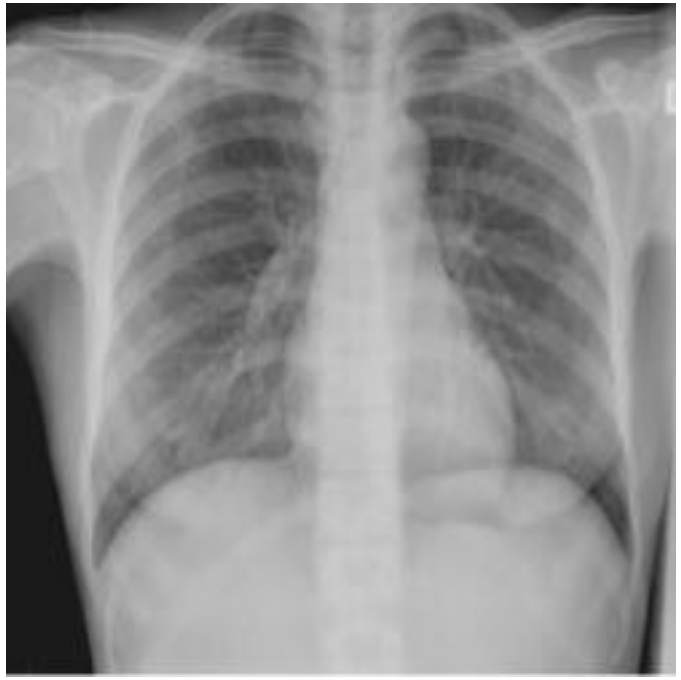
We should reach the following at dome of diaphragm midclavicular line:

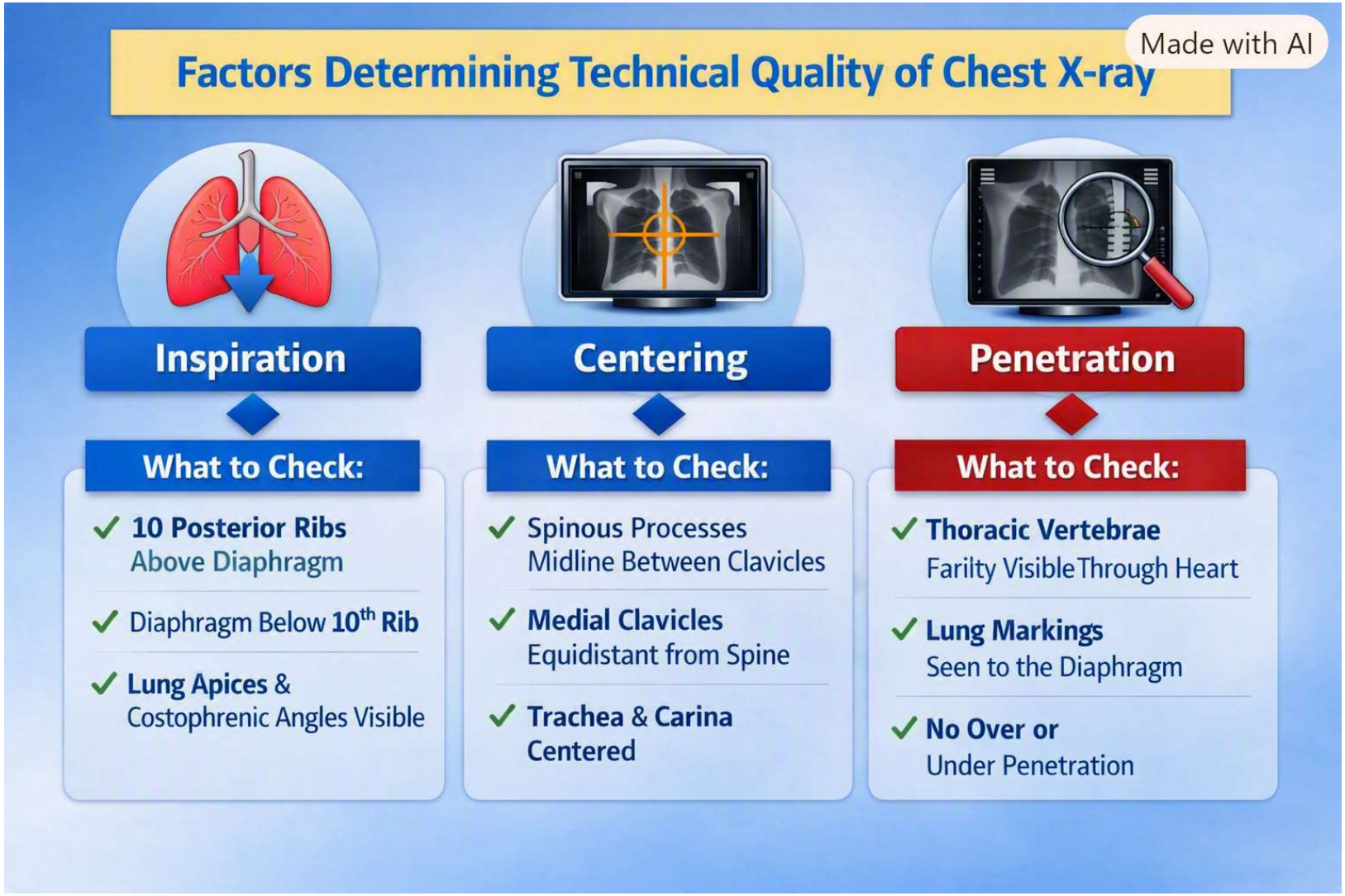
- **6th** rib **anteriorly**
- **8th-10th** rib **posteriorly**



Inspiratory VS Expiratory

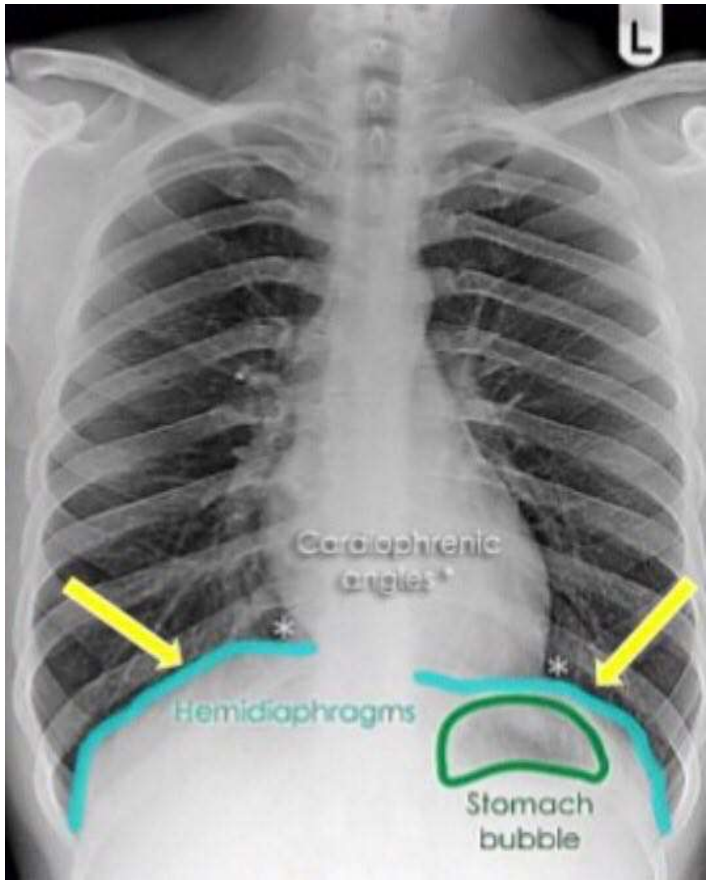


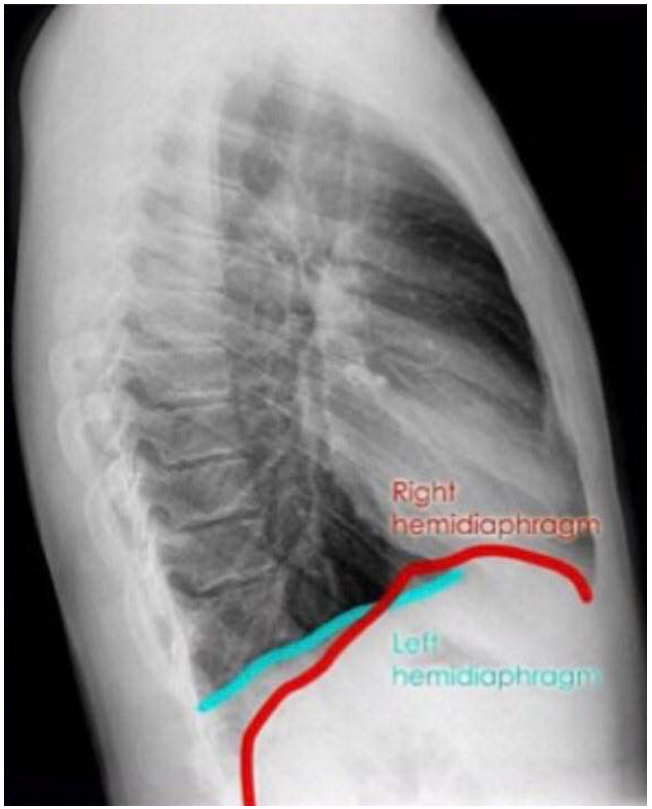



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Factors Determining Technical Quality of Chest X-ray
Inspiration
What to Check:
10 Posterior Ribs
Above Diaphragm
Diaphragm Below 10th Rib
Lung Apices &
Costophrenic Angles Visible
Centering
What to Check:
Spinous Processes
Midline Between Clavicles
Medial Clavicles
Equidistant from Spine
Trachea & Carina
Centered
Penetration
What to Check:
Thoracic Vertebrae
Farilty Visible Through Heart
Lung Markings
Seen to the Diaphragm
No Over or
Under Penetration

Approach to read a chest X ray

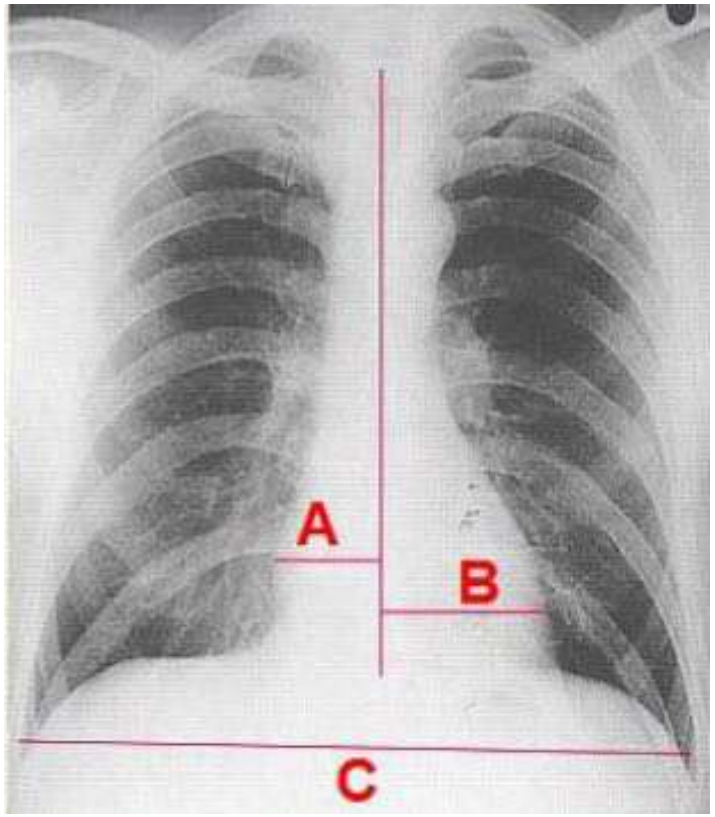
- Trace diaphragm.
- Check size, position and shape of heart.
- Check position and look at mediastinum.
- Examine hilar shadows.
- Examine lungs fissures and zones.
- Look at hidden areas.
- Check pleura and pleural spaces.
- Check bony skeleton.
- Look at soft tissue.

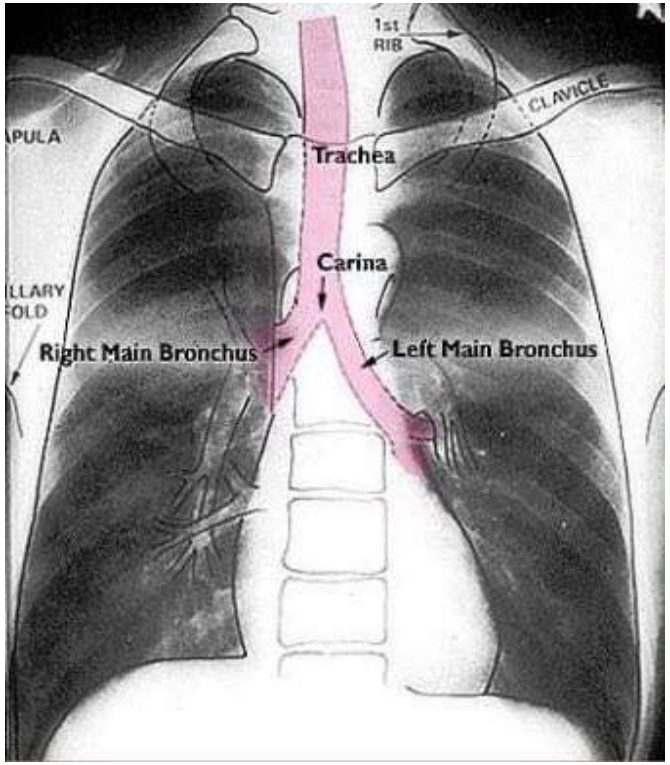

L
Cardiophrenic angles
Hemidiaphragms
Stomach bubble

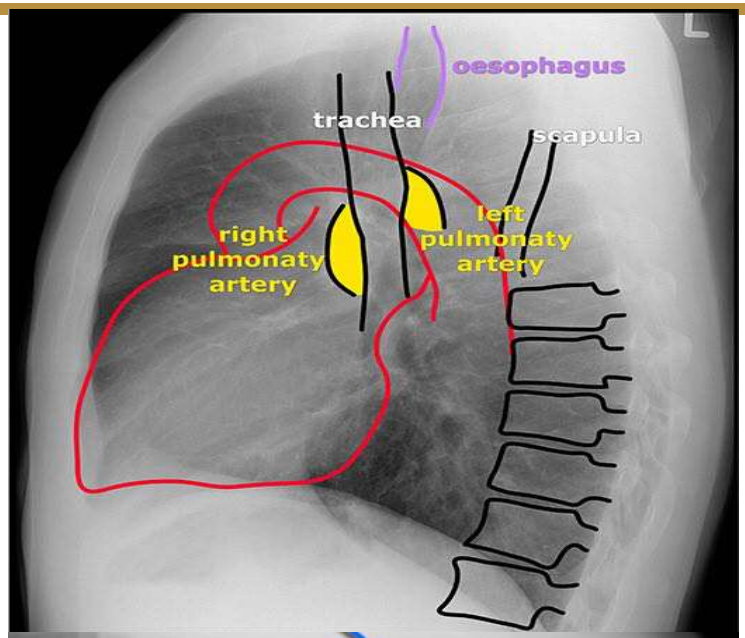

Right hemidiaphragm
Left hemidiaphragm

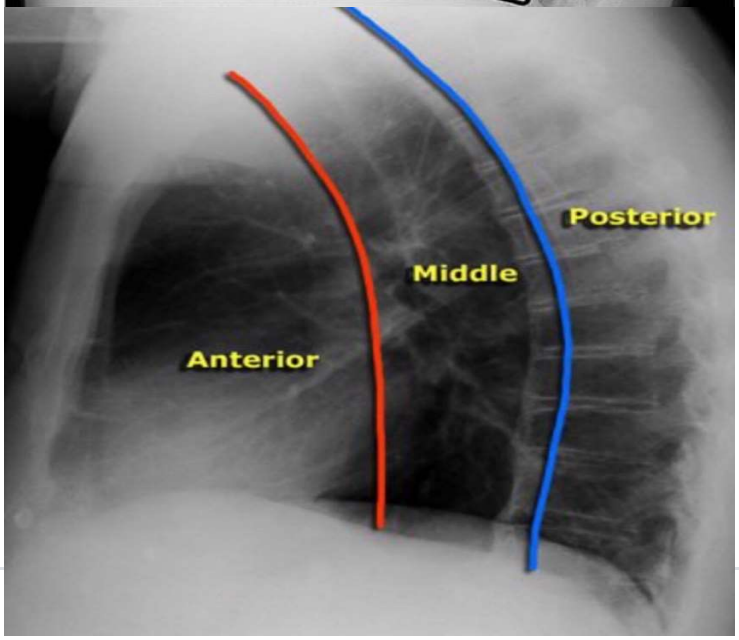
Cardiothoracic ratio:

- The cardiothoracic ratio should be less than **0.5**. i.e. **A+B/C<0.5**
- A cardiothoracic ratio **> 0.5** suggests **cardiomegaly in adults**
- A cardiothoracic ratio **> 0.6** suggests **cardiomegaly in newborn**.




1st RIB
CLAVICLE
Trachea
Carina
Right Main Bronchus
Left Main Bronchus


oesophagus
trachea
scapula
right pulmonaty artery
left pulmonaty artery

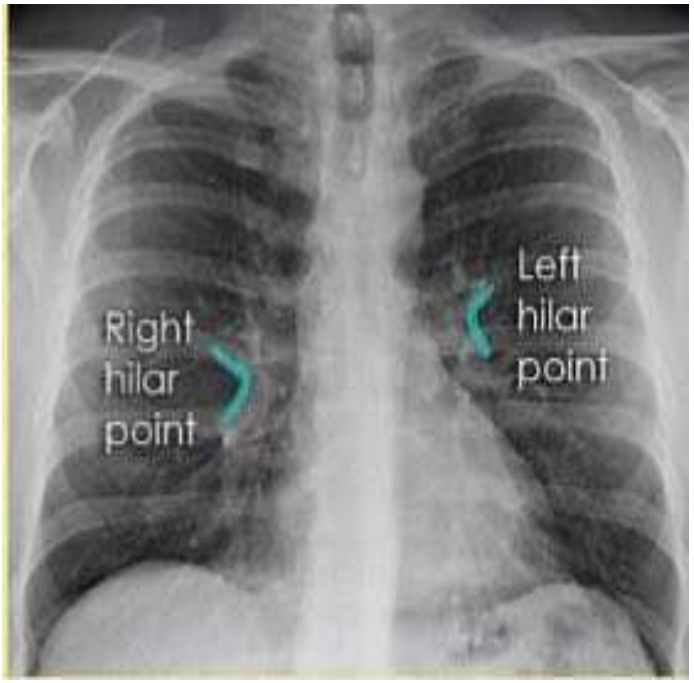

Posterior
Middle
Anterior

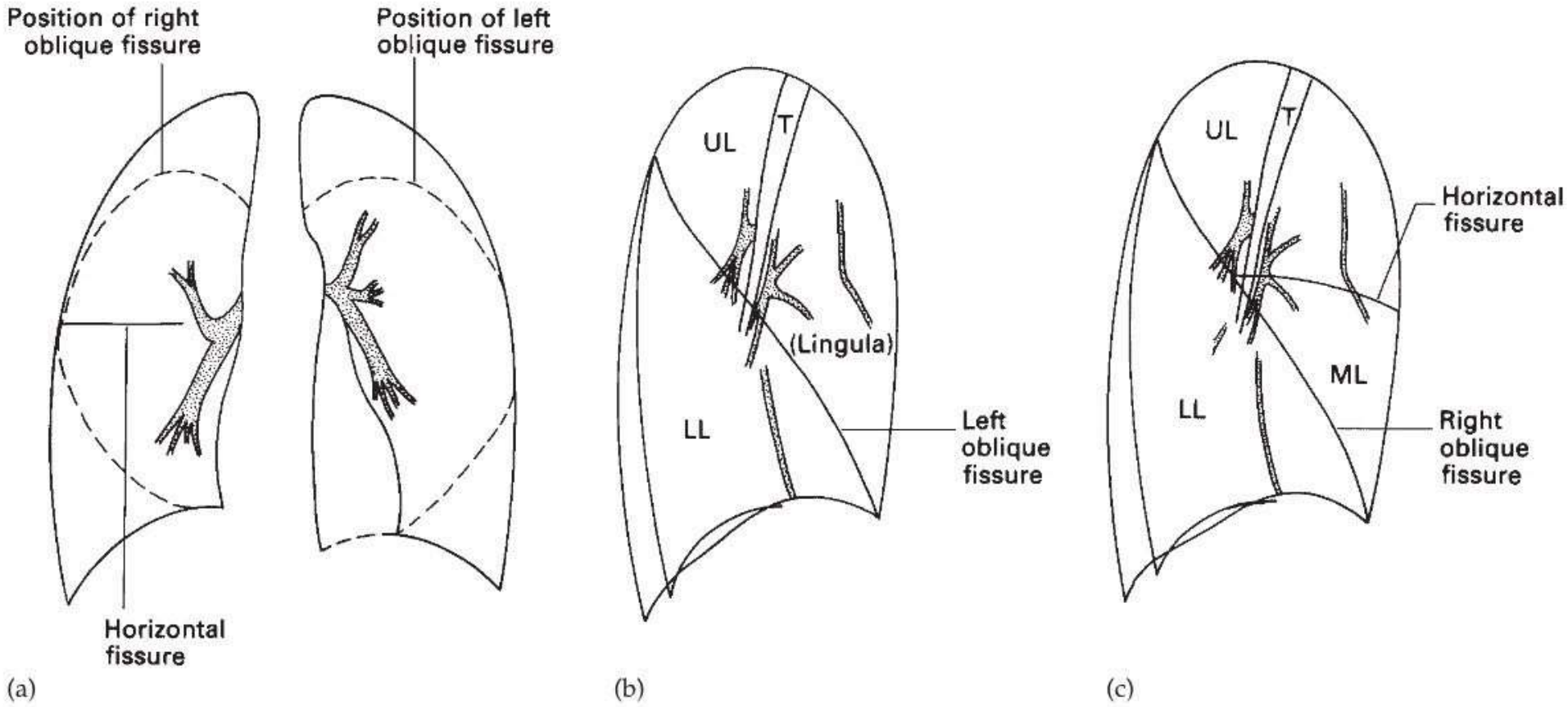
The structures contributing to hilar shadows are:

- **Major:** Pulmonary artery and veins
- **Minor:** Fat, Lymph nodes and Bronchial walls (not visible unless abnormal)

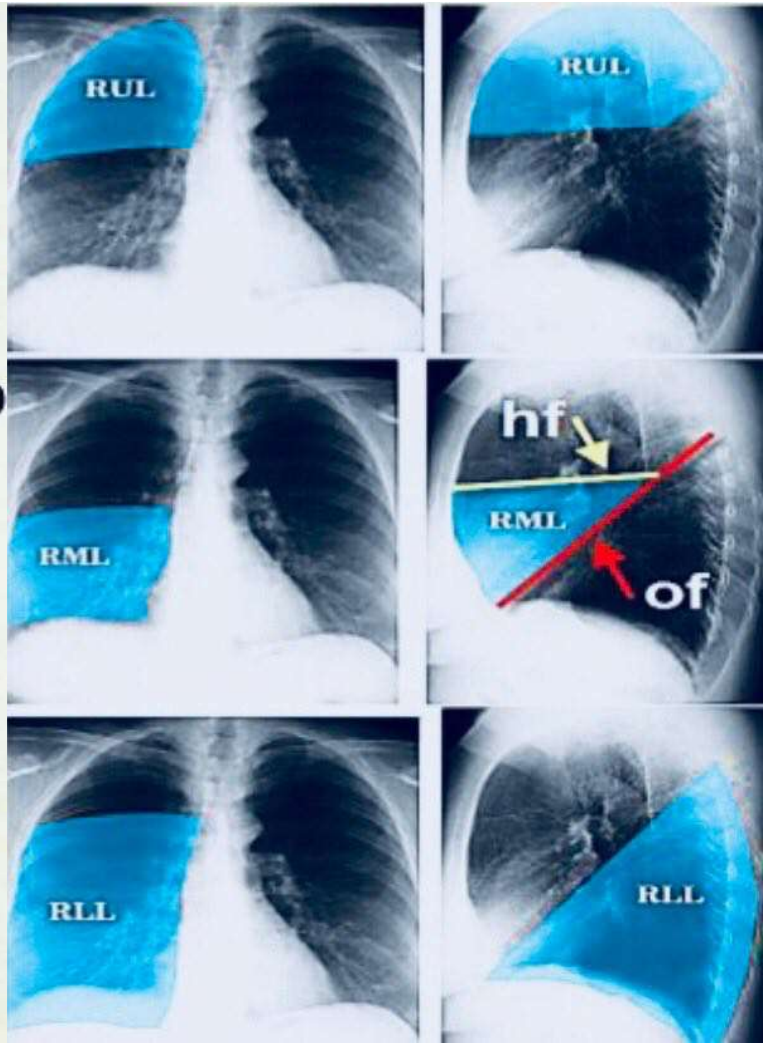
Normal Hilum:

- **Position:** Left hilum is **slightly higher** than the right hilum, Only in a minority of cases the right hilus is at **the same level** as the left, but **never higher**.
- **Shape:** **Concave**
- **Size:** **Similar** on both sides
- **Density:** Almost **same** on both sides

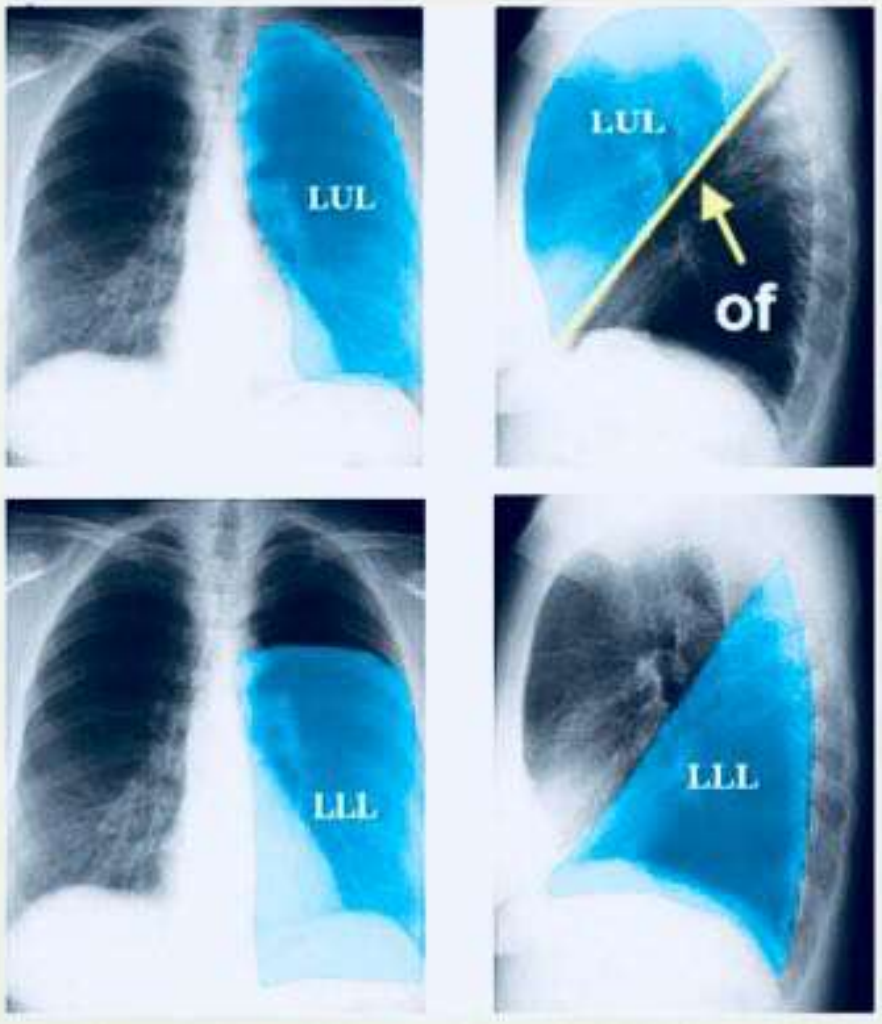



Position of right oblique fissure
Position of left oblique fissure
Horizontal fissure
(a)
UL
T
(Lingula)
LL
Left oblique fissure
(b)
UL
T
Horizontal fissure
ML
LL
Right oblique fissure
(c)

- In **the right** lung there is **an oblique fissure (of)** and **a horizontal fissure (hf)** , separating the lung into three lobes - **upper**, **middle**, and **lower.**
- Each lobe has its own visceral pleural covering.

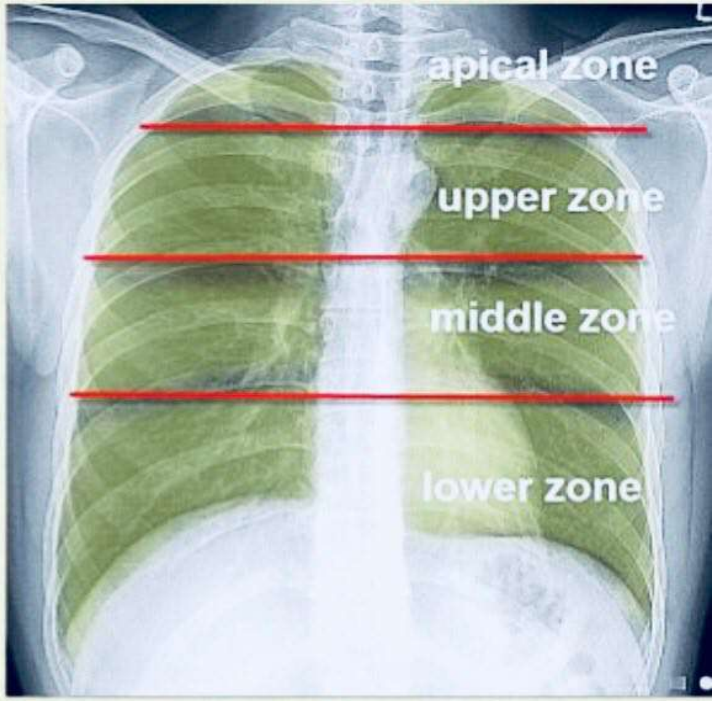


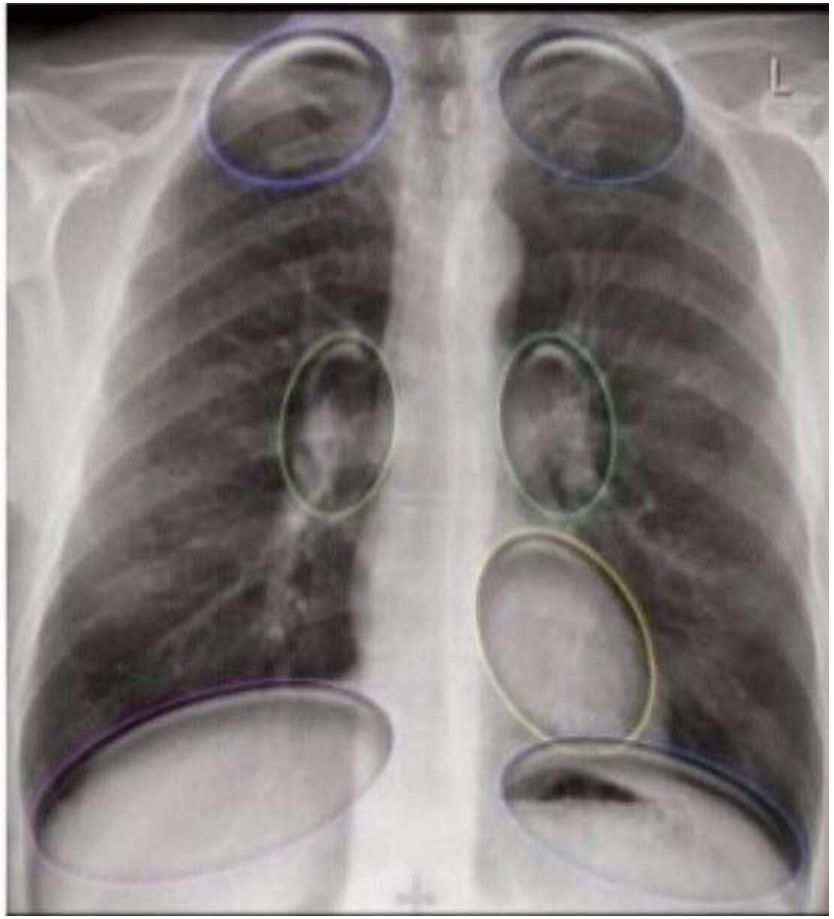
- The **left lung** is divided into **two lobes**, **upper** and **lower**.
- These lobes have their own pleural covering and these lie together to form **the oblique (major) fissure (of)**.

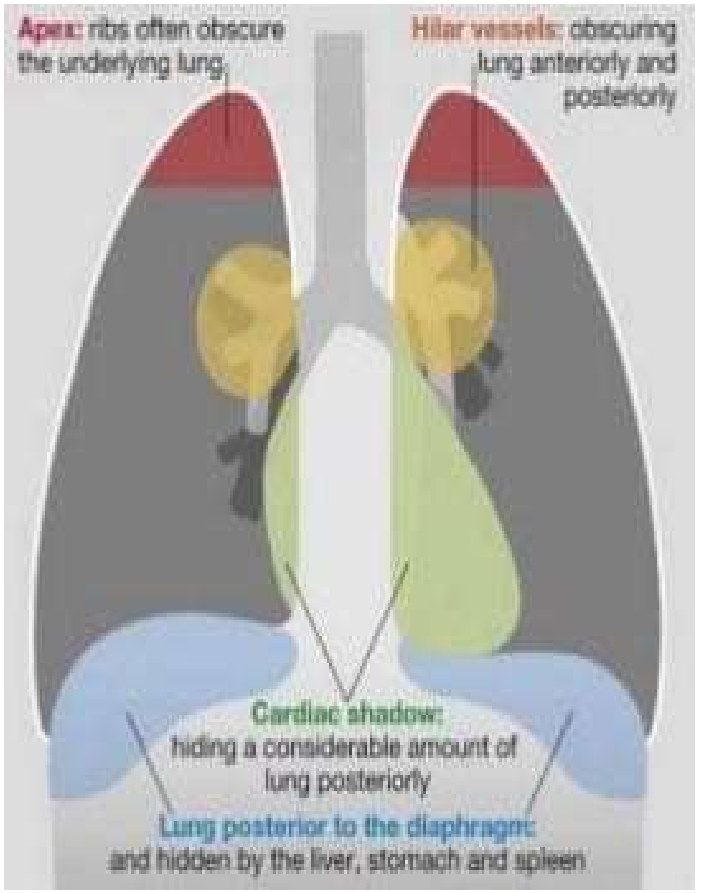


The chest radiograph zones:

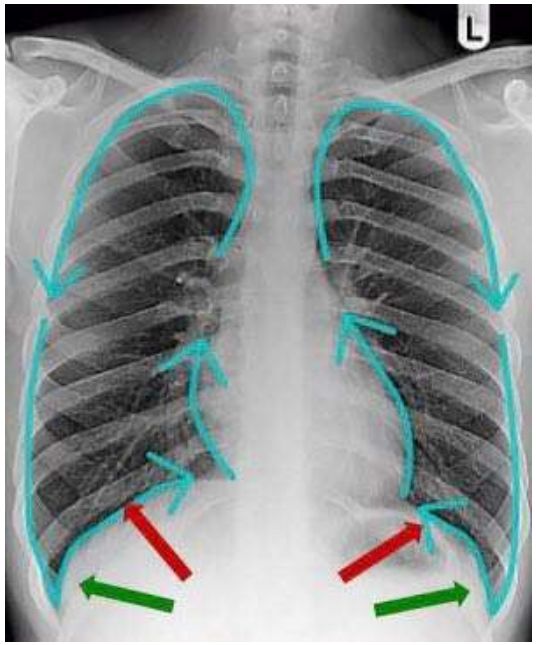
1. **Apical zone:** above the clavicles
2. **Upper zone:** below the clavicles and above the **cardiac silhouette** (i.e. up to lower margin of 2nd rib)
3. **Mid zone:** the level of the hilar structures (i.e. from lower margin of 2nd rib to lower margin of 4th rib)
4. **Lower zone:** the bases (i.e. from 4th rib to diaphragm)

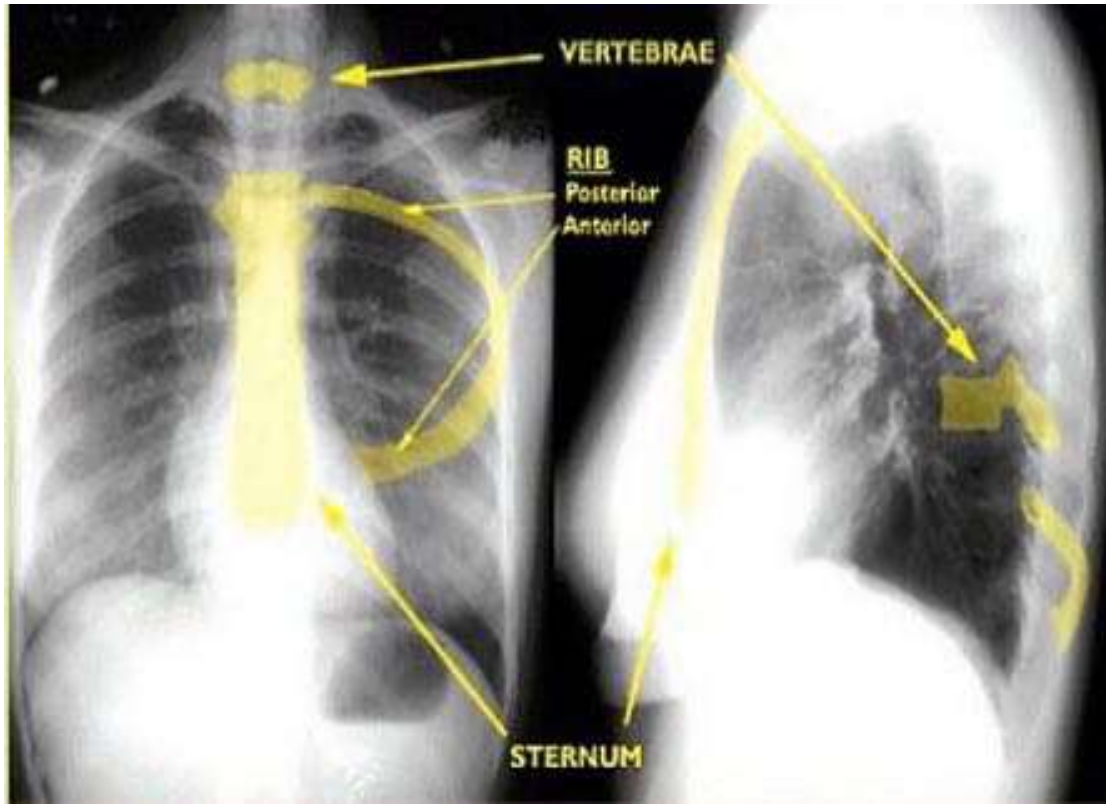




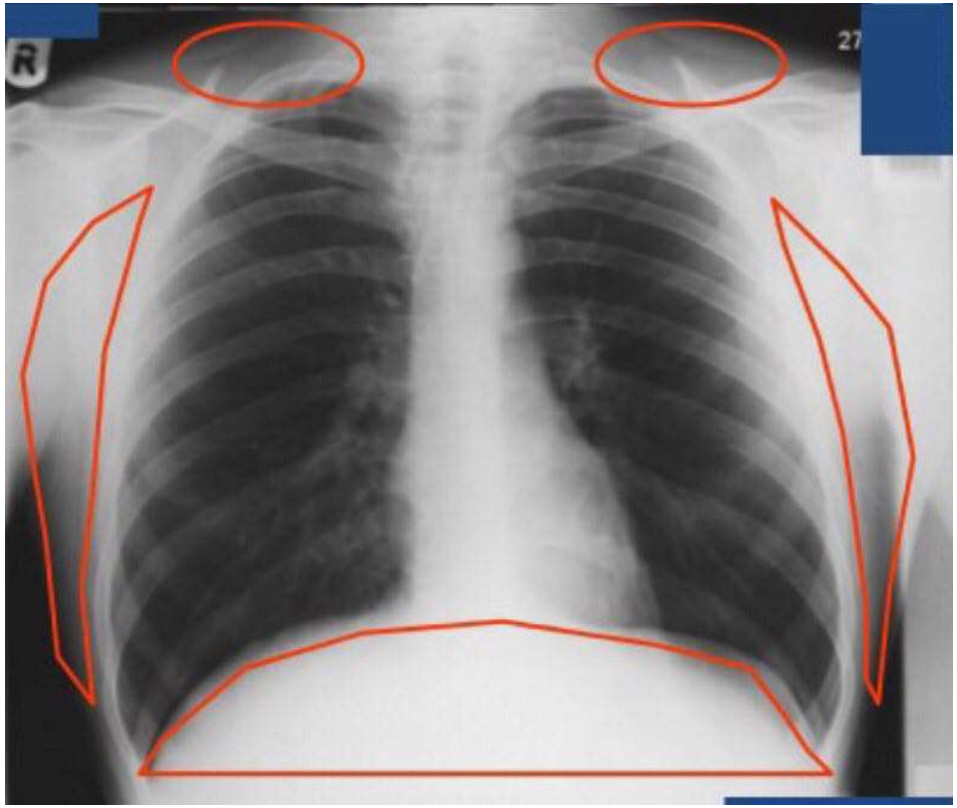

Apex: ribs often obscure the underlying lung
Hilar vessels: obscuring lung anteriorly and posteriorly
Cardiac shadow: hiding a considerable amount of lung posteriorly
Lung posterior to the diaphragm: and hidden by the liver, stomach and spleen

- Trace round the **edge** of the lung. where pleural abnormalities are seen.
- Start and end at **the hila.**
- Is there any **pleural thickening?**
- The **lung markings** should be visible to the **chest wall.**
- Is there **pneumothorax?**
- Is there **pleural effusion?**




VERTEBRAE
RIB
Posterior
Anterior
STERNUM

- Supraclavicular region: (enlarged LN).
- Lateral chest wall: (surgical emphysema)
- Under diaphragm: (pneumoperitoneum)
- Breast shadow: (if there is mastectomy)



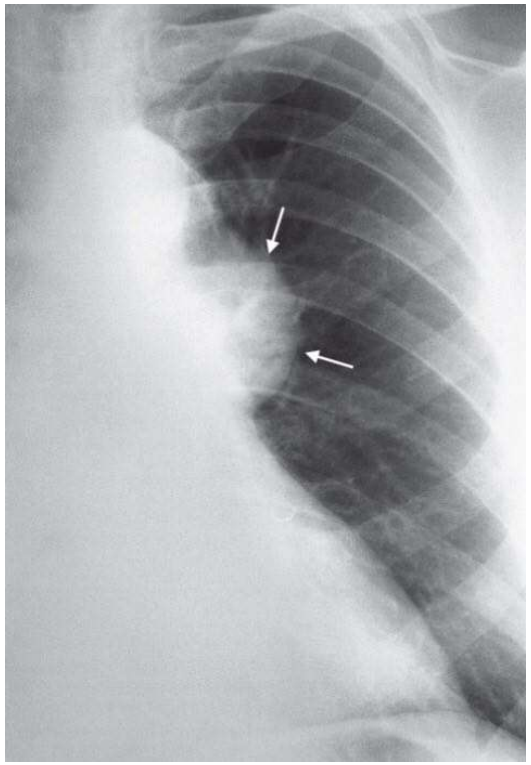
Diseases of the chest with normal X ray:

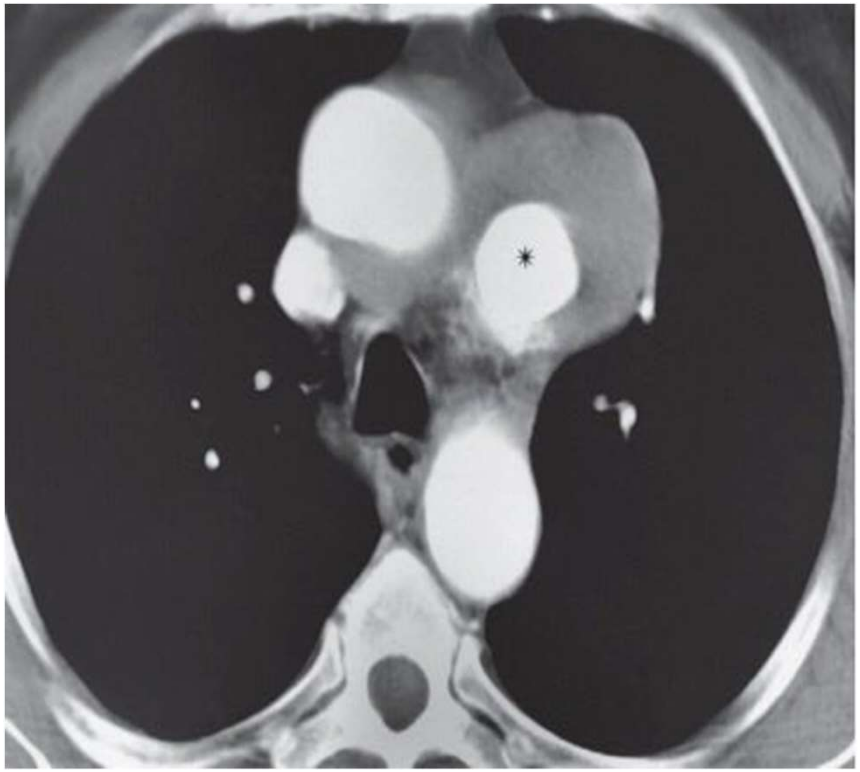
1. **Obstructive airway disease** (asthma and acute bronchiolitis).
2. Early **bronchiectasis.**
3. **Small pulmonary** lesion and **endobronchial** lesion.
4. Pulmonary **emboli without infarction.**
5. Some types of **infection** (early or mild).
6. **Dry pleurisy** and **small pleural effusion**.
7. **Mediastinal masses.**

Main indications of chest CT scan:

1. Diagnosing **diffuse pulmonary disease.**
2. Showing extent and severity of **bronchiectasis**..
3. Allowing **fat** to be recognized (which is useful to diagnose fatty tumor).
4. Showing the **shape of intrapulmonary or pleural** abnormality.
5. Localizing lesion **prior to biopsy.**
6. Showing extent of **mediastinal lesions.**
7. distinguishing **vascular from non-vascular** lesion (e.g. aneurysm from solid mass).
8. Diagnosing **pulmonary emboli** using CT pulmonary angiography.

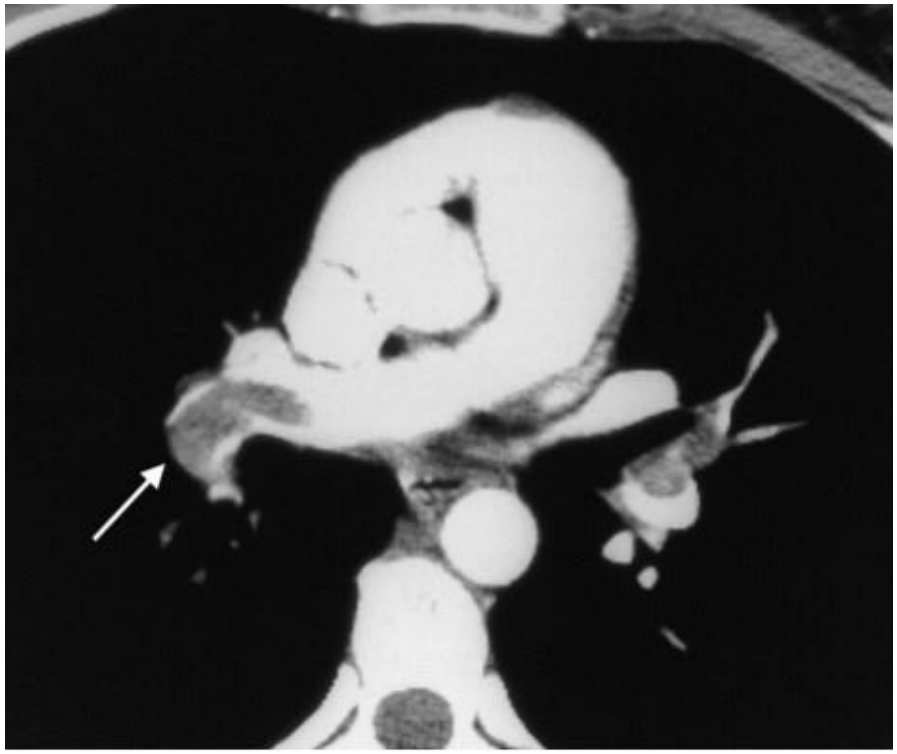
45 year old male presented with chest pain





35 year old male presented with SOB





Radiological signs of lung disease:

1. Air space disease (replacement of alveolar air by fluid or other materials):
   A. Consolidation.
   B. Pulmonary edema.
2. Linear shadows.
3. Nodules and masses.
4. Widespread small shadows.
5. Collapse.

References

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4. Rumack CM, Wilson SR, Charboneau JW, Levine D. Diagnostic Ultrasound. 5th ed. Elsevier; 2018.
5. Fraser RS, Müller NL, Colman N, Paré PD. Diagnosis of Diseases of the Chest. 4th ed. Saunders; 1999.
6. Radiopaedia.org – Open educational radiology resource.

QUESTIONS & DISCUSSION

Thank you

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